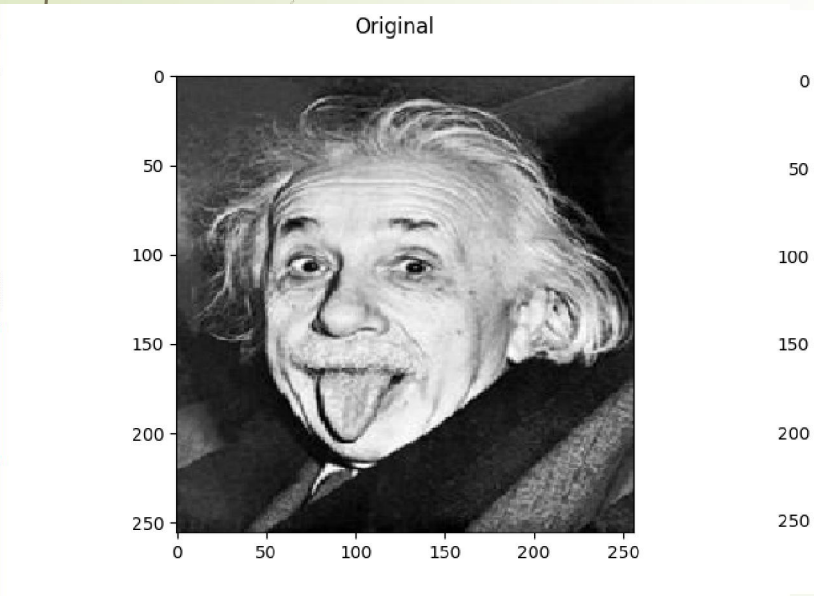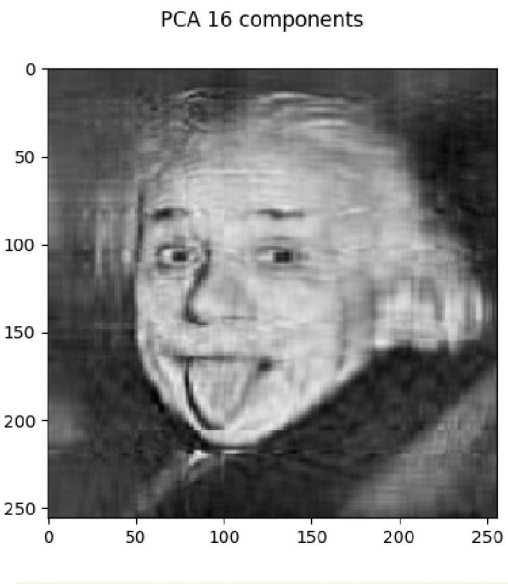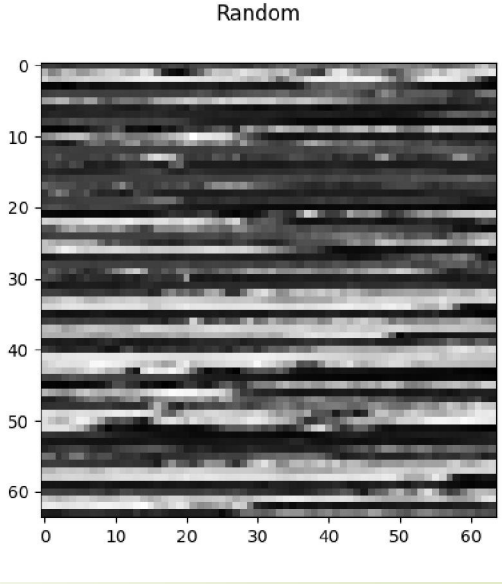I used an Einstein image and applied PCA to reduce the number of dimensions to 16. I also randomly picked 16 shades to compress the image. PCA to reduce dimensions is much better than randomly choosing shades since PCA keeps the most essential shades.